

- PLAN:
- 1) Random Vectors: Defn, Mean, Correlation, Covariance, Examples.
 - 2) Random Processes: Defn, Examples.
Finite Dimensional Distributions: Consistency
Properties: Mean, Autocorrelation, Autocovariance
 - 3) Stationary & WSS Processes

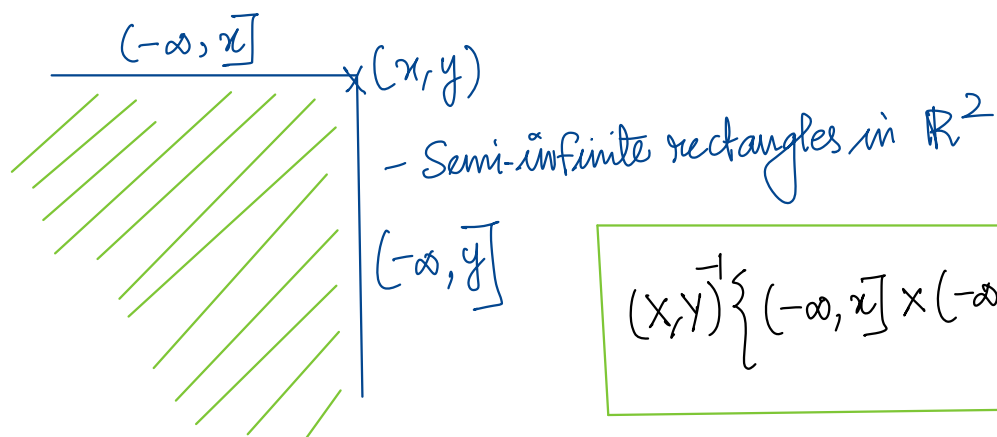
- REF:
- 1) Bruce Hajek, "Random Processes for Engineers"
 - 2) Anurag Kumar, "Discrete Event Stochastic Processes"

Random Vectors:

- Bivariate Random Vector: Fix a probability space $(\Omega, \mathcal{F}, P)$

Defn: Given two functions $X: \Omega \rightarrow \mathbb{R}$ and $Y: \Omega \rightarrow \mathbb{R}$,
we say $(X, Y): \Omega \rightarrow \mathbb{R}^2$ is a **bivariate random vector** w.r.t $\mathcal{F}$ if

$$\{\omega: X(\omega) \leq x\} \cap \{\omega: Y(\omega) \leq y\} \in \mathcal{F} \quad \forall x, y \in \mathbb{R}$$



$$(X, Y)^{-1}\{(-\infty, x] \times (-\infty, y]\} \in \mathcal{F} \quad \forall x, y \in \mathbb{R}$$

Since this must hold $\forall x, y \in \mathbb{R}$

$$\Rightarrow \{\omega: X(\omega) \leq x\} \cap \underbrace{\{\omega: Y(\omega) < \infty\}}_{\Omega} \in \mathcal{F} \quad \forall x \in \mathbb{R}$$

$$\Rightarrow \{\omega: X(\omega) \leq x\} \in \mathcal{F}, \quad \forall x \in \mathbb{R} \quad \Rightarrow X \text{ is a random variable.}$$

Similarly, Y is a random variable.

(X, Y) is a random vector $\Rightarrow X, Y$ are random variables.

Qn: Is the reverse implication true?

i.e., If X and Y are random variables w.r.t $\mathcal{F}$, is (X, Y) a random vector w.r.t $\mathcal{F}$?

Answer: YES!!

Proof: $A, B \in \mathcal{F} \Rightarrow A^c, B^c \in \mathcal{F} \Rightarrow (A^c \cup B^c) \in \mathcal{F}$
 $\Rightarrow (A^c \cup B^c)^c \in \mathcal{F} \Rightarrow A \cap B \in \mathcal{F}$

$\therefore X$ is a RV w.r.t $\mathcal{F} \Rightarrow \{\omega: X(\omega) \leq x\} \in \mathcal{F} \quad \forall x \in \mathbb{R}.$

Y is a RV w.r.t $\mathcal{F} \Rightarrow \{\omega: Y(\omega) \leq y\} \in \mathcal{F} \quad \forall y \in \mathbb{R}.$

Both RVs are defined w.r.t the same $(\Omega, \mathcal{F}, \mathbb{P})$

$\therefore \{\omega: X(\omega) \leq x\} \cap \{\omega: Y(\omega) \leq y\} \in \mathcal{F} \quad \forall x, y \in \mathbb{R}.$

$\Rightarrow (X, Y)$ is a random vector w.r.t $\mathcal{F}$

(X, Y) is a random vector $\Leftrightarrow X, Y$ are random variables.

Going forward, random vectors will be defined as vectors of random variables, all defined on the same probability space!!

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Defn: A **random vector** X w.r.t $\mathcal{F}$ of dimension m has the form $X = [X_1 \ X_2 \ \dots \ X_m]^T$, where X_i s are random variables defined w.r.t $\mathcal{F}$.

Expectation of X : $\mathbb{E}[X] = [\mathbb{E}[X_1] \ \mathbb{E}[X_2] \ \dots \ \mathbb{E}[X_m]]^T$

Auto Correlation of X : $E[XX^T]$ — $m \times m$ correlation matrix.

$$XX^T = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \begin{bmatrix} x_1 & x_2 & \dots & x_m \end{bmatrix} = \begin{bmatrix} x_1^2 & x_1 x_2 & \dots & x_1 x_m \\ x_2 x_1 & x_2^2 & \dots & x_2 x_m \\ \vdots & \vdots & \ddots & \vdots \\ x_m x_1 & \dots & \dots & x_m^2 \end{bmatrix}$$

$$E[XX^T]_{i,j} = E[x_i x_j]$$

Auto Covariance of X: $\text{Cov}(X)$

$$\begin{aligned} \text{Cov}(X)_{i,j} &= \text{Cov}(x_i, x_j) = E[(x_i - E(x_i))(x_j - E(x_j))] \\ &= E[x_i x_j] - E[x_i]E[x_j] \end{aligned}$$

Cross Correlation and Cross Covariance :

Consider two random vectors X of dimension m and Y of dimension n defined w.r.t the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

The cross-correlation between X and Y is the $m \times n$ matrix $E[XY^T]$ with

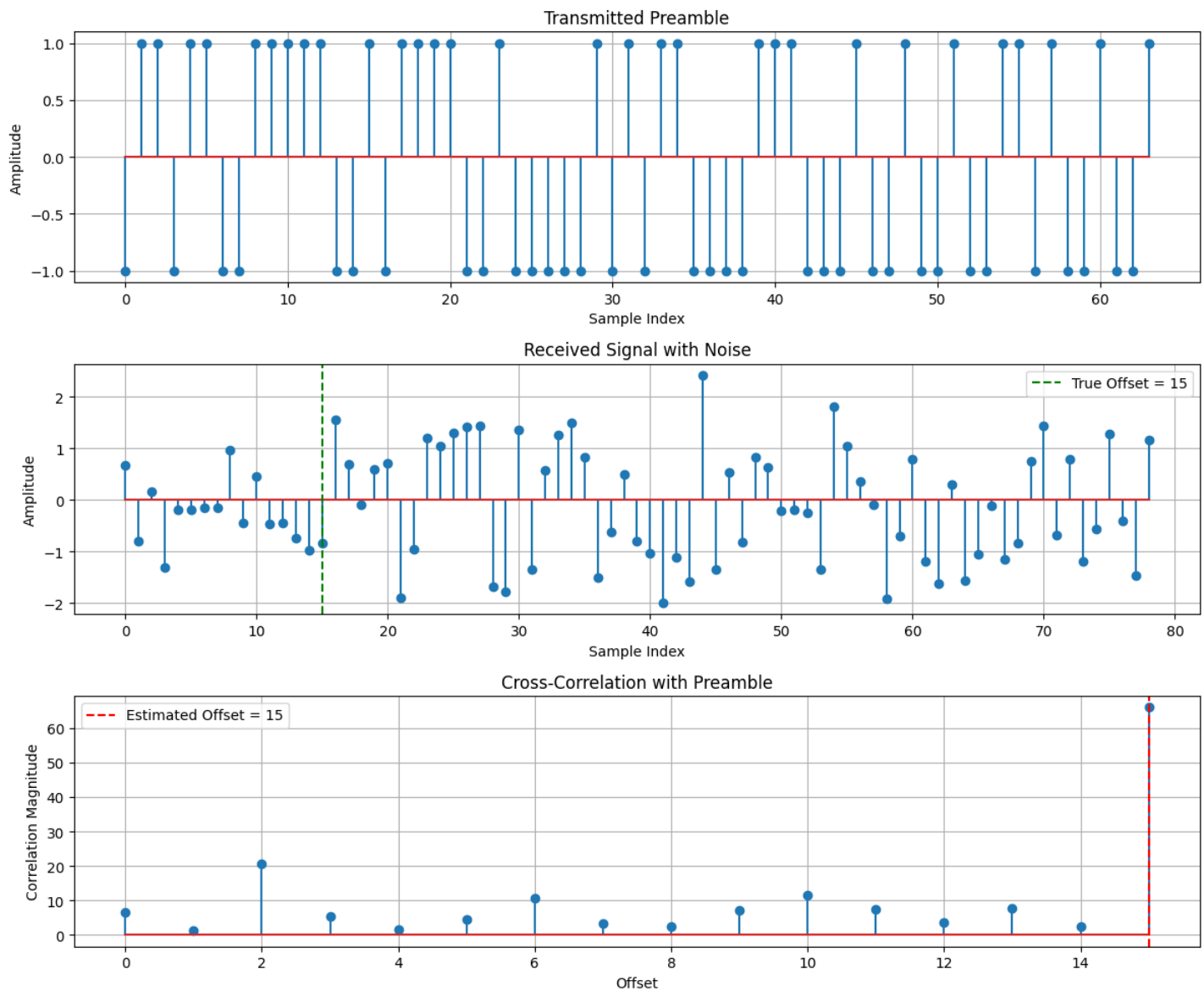
$$E[XY^T]_{i,j} = E[x_i y_j]$$

The cross-covariance between X and Y is the $m \times n$ matrix $\text{Cov}(X, Y)$ given by

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])^T]$$

$$\text{Cov}(X, Y)_{i,j} = \text{Cov}(x_i, y_j).$$

Applications of Cross-Correlation



- 1) Timing Synchronization
 - 2) Signal Detection
 - 3) Channel Estimation
- } In Communication
& Signal Processing

In Computer Vision:

Similarity between two images or image regions. In tasks such as object detection, image alignment, feature/template matching.

A pattern is slid across the given image and similarity scores are computed using cross correlation. For example: Identifying a face in a group photo:

Example 1: $X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0.8 \\ 0.8 & 1 \end{bmatrix}\right)$

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = AX + b, \quad A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \text{ and } b = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Compute $E[Y]$, $\text{Cov}(Y)$ and $\text{Cov}(X, Y)$.

$$E[Y] = AE[X] + b = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\text{Cov}(Y) = A \text{Cov}(X) A^T = \begin{bmatrix} 8.2 & 2.8 \\ 2.8 & 1 \end{bmatrix}$$

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)^T] = \begin{bmatrix} 2.6 & 0.8 \\ 2.8 & 1 \end{bmatrix}$$

Example 2: $X_1 \sim \text{unif}[0, 1]$, $X_2 = X_1 + W$, $W \sim \text{unif}[0, 1]$, $X_1 \perp W$

Let $X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$. Find $E[X]$, $E[XX^T]$ and $\text{Cov}(X)$. (Homework)

Example 3: Consider $X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$, where X_1, X_2 are random variables defined on the same probability space with joint distribution

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} 6x_2, & 0 \leq x_1 \leq 1, 0 \leq x_2 \leq x_1 \\ 0 & \text{otherwise} \end{cases}$$

Find $E[X]$ and $E[XX^T]$

(Homework)

RANDOM PROCESS : Definition

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Defn: A **random process** is a collection of random variables $\{X_t, t \in \mathcal{T}\}$, indexed by the elements of an **index set** $\mathcal{T}$, all defined w.r.t $\mathcal{F}$.

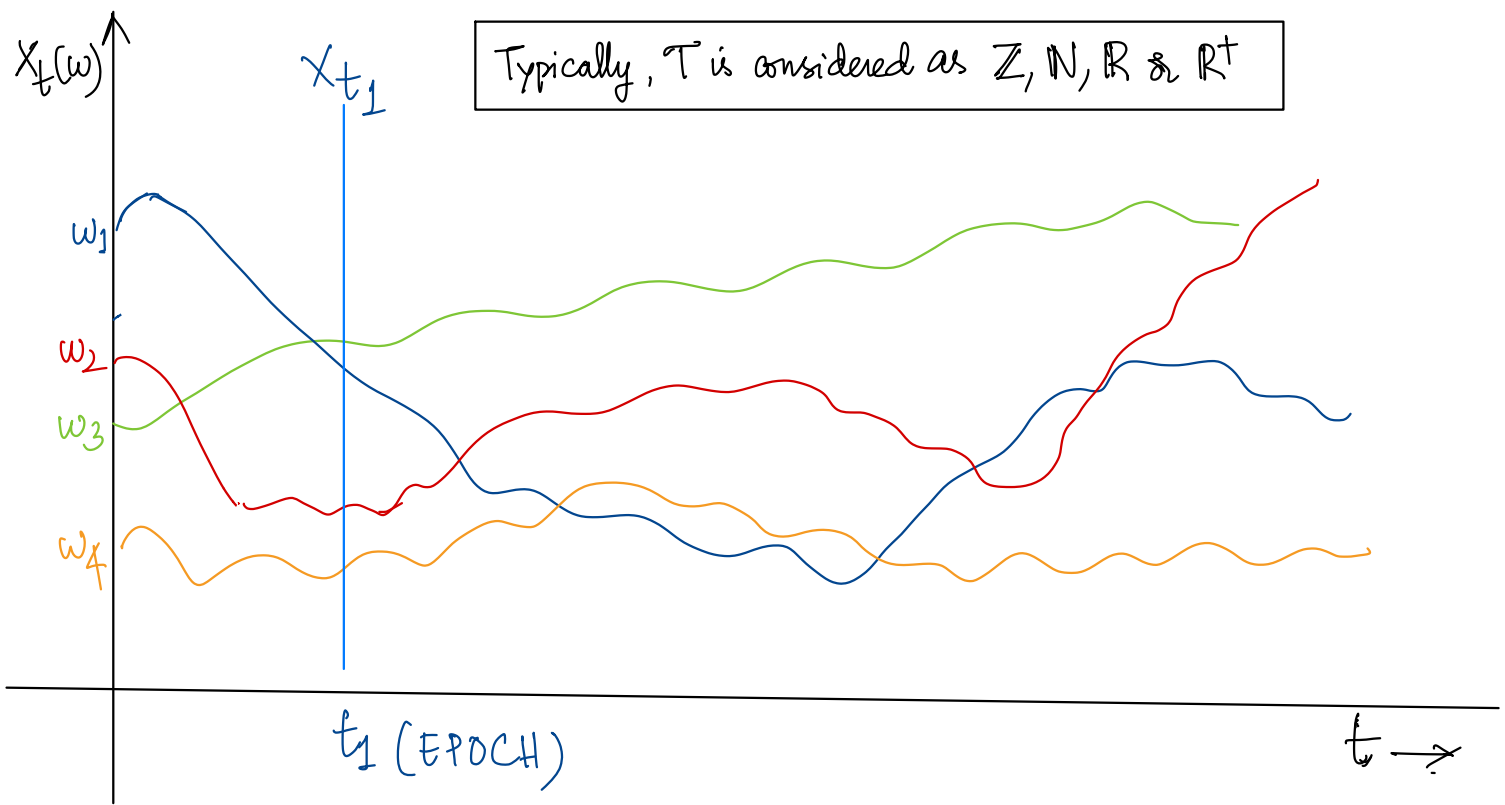
- When $\mathcal{T}$ is finite, say $\mathcal{T} = \{1, 2, \dots, d\}$, we have a random vector $X = [X_1 \dots X_d]^T$
- If $\mathcal{T}$ is a countably infinite set, $\{X_t, t \in \mathcal{T}\}$ is a **discrete parameter** process
- If $\mathcal{T}$ is uncountably infinite $\{X_t, t \in \mathcal{T}\}$ is a **continuous parameter** process.
- Often, $\mathcal{T}$ has the interpretation of **time**.
- Random processes are also called **Stochastic Processes** (from a Greek word for "guess")

Interpretations of a Random Process $\{X_t, t \in \mathcal{T}\}$

$X_t : \Omega \rightarrow \mathbb{R}$ is a random variable for each $t \in \mathcal{T}$

$X(\omega) : \mathcal{T} \rightarrow \mathbb{R}$ is a **sample path** of the process for each $\omega \in \Omega$.

$X : \mathcal{T} \times \Omega \rightarrow \mathbb{R}$ $X_t(\omega)$ or $X(t, \omega)$ is a real number for each $\omega \in \Omega, t \in \mathcal{T}$.
as a function of two variables.

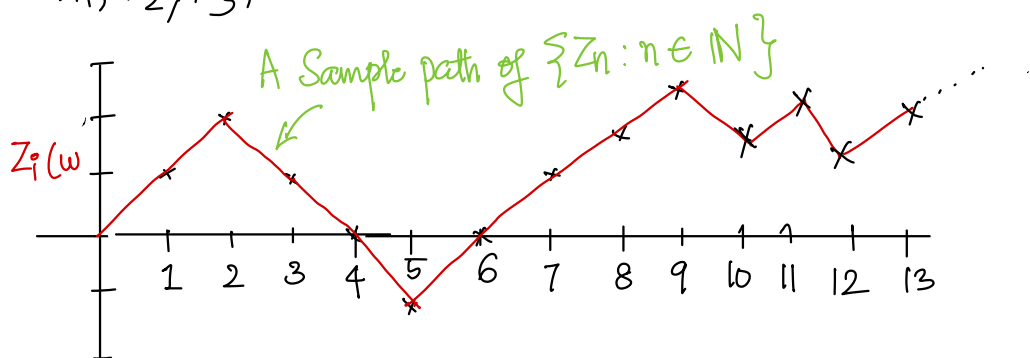


Example: Random Walk.

$\{X_i; i \in \mathbb{N}\}$ is an i.i.d Bernoulli process.

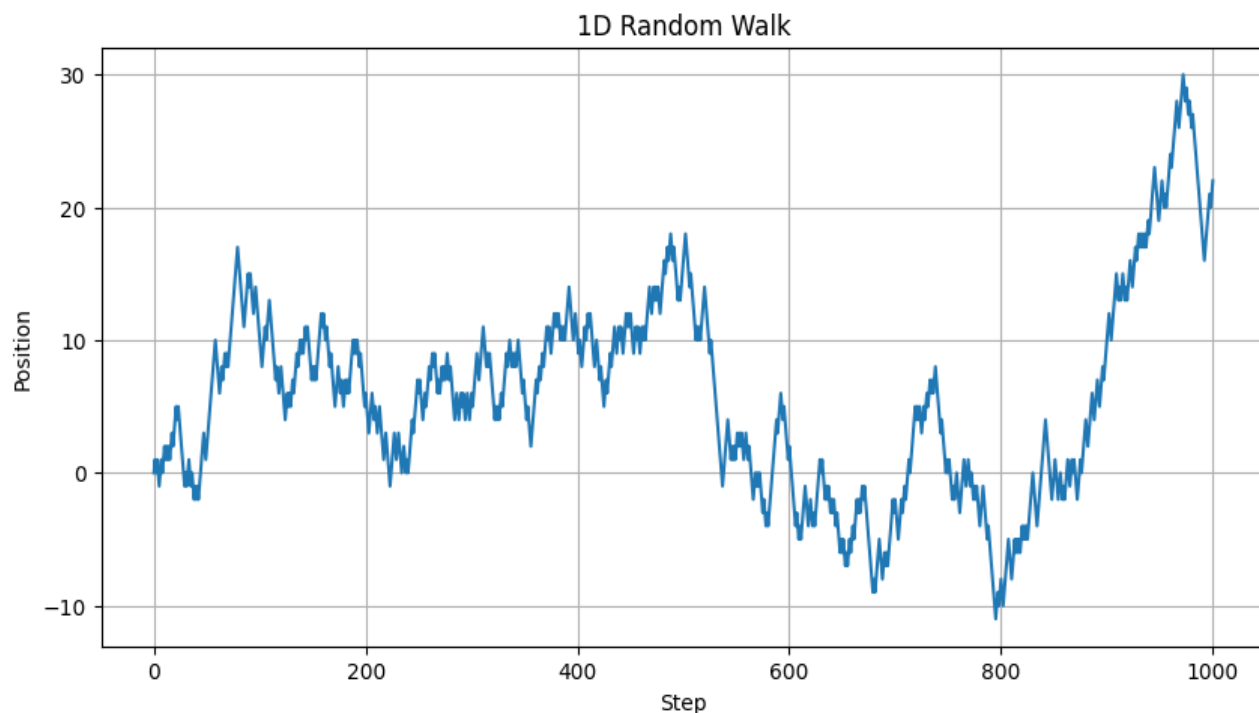
Consider a prob. space $(\Omega, \mathcal{F}, P)$

$X_1, X_2, X_3, \dots$ defined w.r.t $\mathcal{F}$, i.i.d with $P(X_i = 1) = \frac{1}{2}$ and $P(X_i = -1) = \frac{1}{2}$.



$$Z_n = \sum_{i=1}^n X_i, \quad n \in \mathbb{N}$$

$Z_1, Z_2, \dots$ is a random walk.



Finite Dimensional Distributions (FDDs)

Fix a probability space $(\Omega, \mathcal{F}, P)$

Defn: Let $\{X_t, t \in \mathcal{T}\}$ be a random process defined w.r.t $\mathcal{F}$.

For any $m \in \mathbb{N}$ and a finite set $\{t_1, t_2, \dots, t_m\} \subset \mathcal{T}$,

$F_{t_1, t_2, \dots, t_m}(x_1, x_2, \dots, x_m) = P(X_{t_1} \leq x_1, X_{t_2} \leq x_2, \dots, X_{t_m} \leq x_m), \quad (x_1, \dots, x_m) \in \mathbb{R}^m$
is called a **finite dimensional distribution** of the process $\{X_t, t \in \mathcal{T}\}$

The collection of all such finite dimensional distributions (i.e., $\forall m \in \mathbb{N}$, and all finite sets $\{t_1, t_2, \dots, t_m\} \subset \mathcal{T}$) is called the set of FDDs for the process $\{X_t, t \in \mathcal{T}\}$

Remark: FDDs cannot be an arbitrary collection of joint CDFs

For example: If (X, Y) is a random vector, then the joint distribution $F_{X,Y}(x, y)$ and the individual distributions $F_X(x)$ and $F_Y(y)$ cannot be arbitrary

They need to satisfy: $F_X(x) = F_{X,Y}(x, \infty)$ and $F_Y(y) = F_{X,Y}(\infty, y)$.

i.e., the distributions should be **consistent**.

Consistency of FDDs

Defn: The set of FDDs of the process $\{X_t, t \in T\}$ are said to be **consistent** if for any $m, n \in \mathbb{N}$, and subsets $T_m \subset T_n \subset T$ with $|T_m| = m$ and $|T_n| = n$,

$$F_{T_m}(x_1, \dots, x_m) = F_{T_n}(\infty, \dots, \infty, x_1, \infty, \dots, \infty, x_2, \dots, \infty, \dots, \infty, x_m, \infty, \dots, \infty)$$

where the dimensions in T_n that are not in T_m are shown as ∞ in the RHS.

Properties of Random Processes: Mean, Autocorrelation and Autocovariance

Fix a probability space $(\Omega, \mathcal{F}, P)$

Let $\{X_t: t \in T\}$ be a random process defined w.r.t $\mathcal{F}$.

The **mean** of the process $\{X_t: t \in T\}$ is a function $\mu_X: T \rightarrow [-\infty, +\infty]$ defined as

$$\mu_X(t) = \mathbb{E}[X_t], \quad t \in T.$$

ENSEMBLE MEAN

The **autocorrelation** and **autocovariance** of the process $\{X_t: t \in T\}$ are functions

$R_X, C_X: T \times T \rightarrow [-\infty, +\infty]$ defined as

$$R_X(t, s) = \mathbb{E}[X_t X_s],$$

$$C_X(t, s) = \text{cov}(X_t, X_s), \quad t, s \in T.$$

Example: IID Process.

Consider a prob. space $(\Omega, \mathcal{F}, P)$

$X_1, X_2, X_3, \dots$ defined w.r.t $\mathcal{F}$, i.i.d with $P(X_i = 1) = p$ and $P(X_i = -1) = 1-p$

$\{X_i: i \in \mathbb{N}\}$ is an i.i.d Bernoulli process.

$$F_{t_1, t_2, \dots, t_n}(x_1, \dots, x_n) = \prod_{i=1}^n F_{t_i}(x_i) \quad \text{Are they consistent?}$$

$$\mu_X(t) =$$

$$R_X(t, s) =$$

$$C_X(t, s) =$$

Let X_0 be an integer valued random variable on the same prob. space independent of $X_1, X_2, \dots$

We define $Z_n = \sum_{i=0}^n X_i$ $\{Z_n: n \geq 0\}$ is a random walk

$$\mu_Z(n | X_0 = k) =$$

Gambler's Ruin: Gambler's initial wealth is k , $k \geq 0$

Z_n is the number of units of money he has at time n .

Let the gambler has a goal of accumulating b units of money, $b > k$

While the random walk $\{Z_n: n \geq 0\}$ continues forever, we are only interested in it until it hits either 0 (gambler is ruined) or b (the gambler is successful).